

بعداً

High Dim. Statistics : A non-asymptotic view

$$X | Y=1 \sim \mathcal{N}(\mu, \Sigma)$$

$$X | Y=-1 \sim \mathcal{N}(-\mu, \Sigma)$$

$$X \in \mathbb{R}^d, Y \in \{-1, 1\}$$

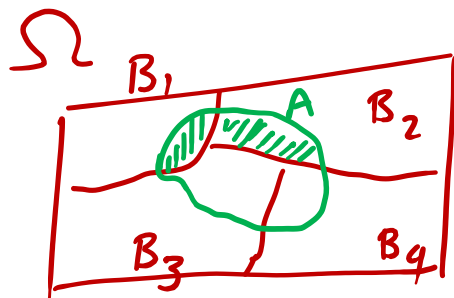
$$\frac{1}{(\sqrt{2\pi})^d \sqrt{\det(\Sigma)}} \exp\left\{-\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu)\right\}$$

Goal : $g(x) \in \{-1, 1\}$

$$\max_g \mathbb{P}(g(x) = Y)$$

$$\min_g \underbrace{\mathbb{P}(g(x) \neq Y)}_{\substack{\text{error} \\ \text{prob. of err.}}}$$

$$\begin{aligned}
 P(g(x) \neq Y) &= P(g(x) \neq Y, \underbrace{Y=1}_{B_1}) \\
 &\quad + P(g(x) \neq Y, \underbrace{Y=-1}_{B_2})
 \end{aligned}$$



$$\begin{aligned}
 P(A) &= P(A \cap B_1) + P(A \cap B_2) \\
 &\quad + \dots + P(A \cap B_n)
 \end{aligned}$$

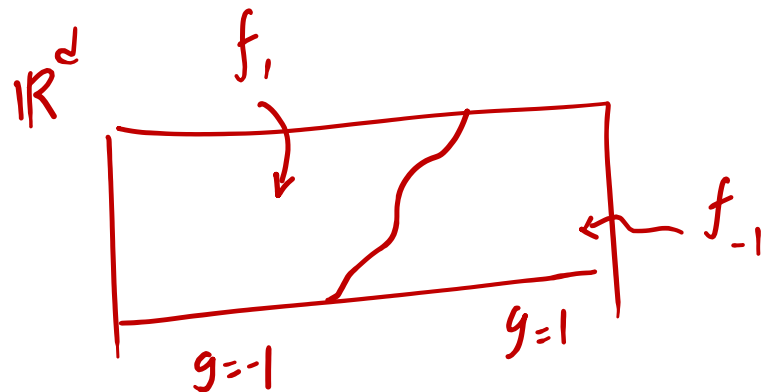
$$P(A \cap B) = P(A|B) \cdot P(B)$$

$$\begin{aligned}
 P(g(x) \neq Y) &= P(g(x) \neq Y | Y=1) \cdot \overbrace{P(Y=1)}^{1/2} + \\
 &\quad P(g(x) \neq Y | Y=-1) \cdot \underbrace{P(Y=-1)}_{1/2}
 \end{aligned}$$

$$P(g(\underline{x}) \neq Y \mid Y=1) = P(g(\underline{x}) \neq 1 \mid Y=1)$$

$$= \int_{g(\underline{x}) = -1} \underbrace{\frac{1}{(\sqrt{2\pi})^d \sqrt{\det \Sigma}}}_{f_1} \exp\left\{-\frac{1}{2}(\underline{x}-\mu)^T \Sigma^{-1}(\underline{x}-\mu)\right\} d\underline{x}$$

$$P(g(\underline{x}) \neq Y \mid Y=-1) = \int_{g(\underline{x}) = 1} \underbrace{\frac{1}{(\sqrt{2\pi})^d \sqrt{\det \Sigma}}}_{f_{-1}} \exp\left\{-\frac{1}{2}(\underline{x}+\mu)^T \Sigma^{-1}(\underline{x}+\mu)\right\} d\underline{x}$$



$$g = \begin{cases} 1, \\ -1, \end{cases}$$

$$f_1 \geq f_{-1}$$

$$f_1 < f_{-1}$$

$$g(\underline{x}) = \begin{cases} 1 \\ -1 \end{cases}$$

$$\frac{1}{(\sqrt{2\pi})^d \sqrt{\det \Sigma}} \exp \left\{ -\frac{1}{2} (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) \right\}$$

otherwise

$$\geq \frac{1}{(\sqrt{2\pi})^d \sqrt{\det \Sigma}} \exp \left\{ -\frac{1}{2} (\underline{x} + \underline{\mu})^T \Sigma^{-1} (\underline{x} + \underline{\mu}) \right\}$$

$$= \begin{cases} 1 \\ \end{cases}$$

$$(\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) \leq (\underline{x} + \underline{\mu})^T \Sigma^{-1} (\underline{x} + \underline{\mu})$$

$$\cancel{x^T \Sigma^{-1} x} - 2 \underline{\mu}^T \Sigma^{-1} \underline{x} \leq \cancel{x^T \Sigma^{-1} x} + 2 \underline{\mu}^T \Sigma^{-1} \underline{x} \\ + \cancel{\underline{\mu}^T \Sigma^{-1} \underline{\mu}} \quad \quad \quad + \cancel{\underline{\mu}^T \Sigma^{-1} \underline{\mu}}$$

$$\Rightarrow \underbrace{4 \underline{\mu}^T \Sigma^{-1} \underline{x}}_0 \geq 0$$

$$4 \underline{x}^T \Sigma^{-1} \underline{\mu} \geq 0$$

$$\underline{X} \sim \mathcal{N}(\mu, \Sigma) \quad \underline{w}^T \underline{X} \sim \mathcal{N}(w^T \mu, w^T \Sigma w)$$

$$\mathbb{E}\{\underline{w}^T \underline{X}\} = w^T \mathbb{E}\{\underline{X}\} = w^T \mu$$

$$\mathbb{V}\{\underline{w}^T \underline{X}\} = \mathbb{E}\{(w^T \underline{X} - w^T \mu)^2\} = \mathbb{E}\{(w^T (\underline{X} - \mu))^2\}$$

$$= \mathbb{E}\{w^T (\underline{X} - \mu) (\underline{X} - \mu)^T w\}$$

$$= w^T \underbrace{\mathbb{E}\{(\underline{X} - \mu)(\underline{X} - \mu)^T\}}_{\Sigma} w = w^T \Sigma w$$

$$U = \underbrace{\mu^T \Sigma^{-1}}_{w^T} X$$

$$U \sim \mathcal{N}(\underbrace{\mu^T \Sigma^{-1} \mu}_{\mu^T \Sigma^{-1} \mu}, \underbrace{\mu^T \Sigma^{-1} \cancel{\Sigma} \Sigma^{-1} \mu}_I)$$

$| \gamma = 1$

$$U \mid Y = -1 \sim \mathcal{N}(-\mu^T \Sigma^{-1} \mu, \mu^T \Sigma^{-1} \mu)$$

$$\begin{aligned} P(\text{Error}) &= P(U \geq 0 \mid Y = -1) P(Y = -1) \\ &\quad + P(U < 0 \mid Y = 1) P(Y = 1) \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \int_{-\infty}^0 \mathcal{N}_u(\mu^T \Sigma \mu, \mu^T \Sigma \mu) du \\ &\quad + \frac{1}{2} \int_0^{+\infty} \mathcal{N}_u(-\mu^T \Sigma \mu, \mu^T \Sigma \mu) du \end{aligned}$$

$$\Phi(u) = \int_{-\infty}^u \mathcal{N}_u(0, 1) du$$

$$P(\text{Error}) = \frac{1}{2} \Phi\left(\frac{0 - \mu^T \Sigma^{-1} \mu}{\sqrt{\mu^T \Sigma^{-1} \mu}}\right) + \frac{1}{2} \Phi\left(\frac{0 - \mu^T \Sigma^{-1} \mu}{\sqrt{\mu^T \Sigma^{-1} \mu}}\right)$$

$$= \Phi\left(-\frac{\mu^T \Sigma^{-1} \mu}{\sqrt{\mu^T \Sigma^{-1} \mu}}\right) = \Phi\left(-\sqrt{\mu^T \Sigma^{-1} \mu}\right)$$

Statistics

μ unknown

Σ known

$$\begin{pmatrix} X_1 \\ Y_1 \end{pmatrix}, \begin{pmatrix} X_2 \\ Y_2 \end{pmatrix}, \dots, \begin{pmatrix} X_n \\ Y_n \end{pmatrix}$$

iid
 $\sim P_{X,Y}$

$$g(\underline{x}) = \begin{cases} 1 \\ \neq 1 \end{cases}$$

$$\underline{x}^T \Sigma^{-1} \mu \geq 0$$

otherwise

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n Y_i X_i$$

$$\hat{g}(\underline{x}) = \begin{cases} 1 & \underbrace{x^T \Sigma^{-1} \hat{\mu}}_U \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

$$U | Y=1 \underset{\mathcal{D}}{\sim} \mathcal{N}(\mu^T \Sigma^{-1} \hat{\mu}, \hat{\mu}^T \Sigma^{-1} \hat{\mu})$$

$$U | Y=-1 \underset{\mathcal{D}}{\sim} \mathcal{N}(-\mu^T \Sigma^{-1} \hat{\mu}, \hat{\mu}^T \Sigma^{-1} \hat{\mu})$$

$$IP(\text{Error} | \mathcal{D}) = \Phi\left(-\frac{\mu^T \Sigma^{-1} \hat{\mu}}{\sqrt{\hat{\mu}^T \Sigma^{-1} \hat{\mu}}}\right)$$

Convergence

$$Z_n \sim \mathcal{N}(0, 1 + 1/n)$$

$$Z_n \xrightarrow[n \rightarrow \infty]{P} Z \quad : \quad \forall \epsilon > 0, \quad \lim_{n \rightarrow \infty} P(|Z_n - Z| \geq \epsilon) = 0$$

$$\forall \epsilon > 0, \quad \lim_{n \rightarrow \infty} P(|Z_n - Z| < \epsilon) = 1$$

$$Z_n \sim \mathcal{N}(0, 1/n)$$

$$Z_n \xrightarrow[n \rightarrow \infty]{P} 0$$

$$0 \leq \underbrace{P(|Z_n - \overset{0}{\underset{\downarrow}{\mu_z}}| \geq \epsilon)}_{\rightarrow 0} \leq \underbrace{\frac{V(Z)}{\epsilon^2}}_{\rightarrow 0}$$

Continuous Convergence Theorem

$$\begin{array}{l} h \text{ Continuous} \\ Z_n \xrightarrow{P} Z \end{array} \Rightarrow h(Z_n) \xrightarrow[n \rightarrow \infty]{P} h(Z)$$

$$P(\text{Error} | D) = \Phi\left(-\frac{\overbrace{\mu^T \Sigma^{-1} \hat{\mu}}^A}{\underbrace{\sqrt{\hat{\mu}^T \Sigma^{-1} \hat{\mu}}}_B}\right) = \Phi\left(-\frac{A}{\sqrt{B}}\right) \quad \square$$

$$\hat{\mu} = \frac{1}{n} \sum Y_i X_i \sim \mathcal{N}(\mu, \frac{1}{n} \Sigma)$$

$$\begin{aligned} \Rightarrow \hat{\mu} &= \mu + \frac{1}{\sqrt{n}} Z \\ Z &\sim \mathcal{N}(0, \Sigma) \end{aligned}$$

$$u := \sum^{-1/2} \mu^{\leftarrow}$$

$$V := \sum^{-1/2} \hat{\mu}$$

$$\begin{cases} A = u^T V \\ B = v^T V \end{cases}$$

$$\underbrace{\sum^{-1/2} \hat{\mu}}_v = \underbrace{\sum^{-1/2} \mu}_u + \frac{1}{\sqrt{n}} \underbrace{\sum^{-1/2} \mathcal{E}}_{\sim \mathcal{N}(0, I)} \underbrace{\quad}_{z'}$$

$$u^T V = \|u\|^2 + \frac{1}{\sqrt{n}} u^T \mathcal{E}'$$

$$v^T V = \|u\|^2 + \frac{1}{n} \|z'\|^2 + \frac{2}{\sqrt{n}} z'^T \sum^{-1/2} \mu$$

$$P(\text{Error } | D) = \Phi\left(\frac{-A}{\sqrt{B}}\right) =$$

$$\Phi\left(\frac{-\left(\underbrace{\|u\|^2}_Q + \frac{1}{\sqrt{n}} u^T z'\right)}{\sqrt{\underbrace{\|u\|^2}_Q + \underbrace{\frac{1}{n} \|z'\|^2}_{d/n} + \frac{2}{\sqrt{n}} z'^T \Sigma^{-1/2} \mu}}\right)$$

$$\|u\|^2 = \underbrace{\mu^T \Sigma^{-1} \mu}_Q$$

$$\Phi\left(-\sqrt{\underbrace{\mu^T \Sigma^{-1} \mu}_Q}\right)$$

$$\lim_{n \rightarrow \infty} d/n = \alpha$$